Algorithm: Real-Time Face Detection using OpenCV

Step 1: Import the OpenCV Library

Begin by importing the cv2 module, which provides all the necessary functions for image processing, object detection, and video analysis.

Step 2: Load the Pre-trained Haar Cascade Classifier

Load the pre-trained Haar Cascade XML file for face detection using the CascadeClassifier() method. This XML file contains the trained model data required to identify human faces based on Haar features.

Step 3: Initialize the Webcam

Access the system's default webcam using cv2.VideoCapture(0). The argument 0 specifies the primary camera device. This object will continuously capture live frames from the camera.

Step 4: Capture Frames in Real Time

Start an infinite loop to continuously read video frames from the webcam. The read() method returns two values: success (a Boolean indicating whether the frame was captured successfully) and frames (the actual image frame retrieved from the camera).

Step 5: Convert the Frame to Grayscale

For efficient computation, convert each frame from color (BGR) to grayscale. The Haar Cascade algorithm operates faster and more accurately on single-channel grayscale images.

Step 6: Detect Faces

Use the detectMultiScale() method on the grayscale frame to detect one or multiple faces. This method returns the coordinates (x, y) and the dimensions (w, h) for each detected face.

Step 7: Draw Rectangles Around Faces

For each detected face, draw a green rectangular boundary around it using the rectangle() method. This visually highlights the face detection results on the live video feed.

Step 8: Display the Output Frame

Display the processed frame with detected faces in a new window using the imshow() function. The window will refresh continuously, showing real-time detection results.

Step 9: Wait for User Input to Exit

Use the waitKey(1) function to listen for keyboard input. If the user presses the ‘q’ key, the loop will terminate, stopping the face detection process.

Step 10: Release Resources

After exiting the loop, release the webcam and close all OpenCV windows to free system resources.

Summary of the Algorithm

1. Import the OpenCV library.
2. Load the pre-trained Haar Cascade XML file.
3. Initialize the webcam to capture real-time video frames.
4. Convert each captured frame to grayscale.
5. Apply the Haar Cascade classifier to detect faces.
6. Draw bounding rectangles around detected faces.
7. Display the processed video frames in real time.
8. Allow the user to terminate the process by pressing ‘q’.
9. Release the webcam and close all display windows.